Recoverable Readout Mismatch in Smooth-to-Spiking EEG Decoding

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Abstract

Does near-chance performance after conversion reflect lost EEG information or a failed readout? On BCI IV-2a (nine participants), we train and select on session 1 and test on session 2. With the original architecture, converted heads yield 25.27–25.91% balanced accuracy (BA); source-only linear readouts on frozen spikes yield 38.69–42.37%, without target labels or encoder updates. Spiking probes trail smooth probes by 1.30–2.25 percentage points (pp). A post-study second encoder recovers 10.11 pp [5.86, 14.29] through refitting. Sharpness-aware surrogate training (SAST) [1, 2] does not demonstrate benefit over compute-matched ordinary training (-0.42 pp, 95% CI $[-2.88, 2.67]$). Source-only refitting diagnoses readout mismatch; equivalence and online performance are not established.

1 Methods

Data and separation. BCI Competition IV 2a has nine participants, four motor-imagery classes, and 22 EEG channels [3]. Participant-specific models use session 1 runs 1–5 for training (240 trials), run 6 for validation (48), and all of session 2 for testing (288). We retain artifact-marked trials, filter each run at 4–38 Hz, and extract 0–4 s after the cue at 125 Hz. Channel normalization, checkpoint selection, and all main-analysis tuning use source training/validation data only.

Training and conversion. A temporal–spatial convolutional encoder has two leaky integrate-and-fire blocks (16/32 channels, leak 0.9) and a time-averaged linear head. Smooth training uses $\phi(v) = \frac{1}{2} + \arctan(v)/\pi$ for threshold-centered v; raw conversion uses $\mathbf{1}[v \geq 0]$ with weights, threshold 1, and previous-activity reset fixed. Ordinary training, its $2\times$-update control, same-batch SAM [4], and independent-batch SAST each have four source-validation tuning candidates and five paired seeds. AdamW runs 800 updates (1,600 for the matched control); SAM/SAST use two gradient evaluations per update. Direct hard-forward training shares the encoder; EEGNet-8,2 [5] is a reference under the same 800-update budget. Calibration selects one shared threshold from eight source-validation candidates over 0.25–3. A post-study filter-bank common spatial pattern (FBCSP)-style reference [6] uses nine bands within 4–38 Hz, one-vs-rest spatial filters, mutual-information feature selection, and shrinkage linear discriminant analysis; four source-validation settings, one deterministic fit per participant.

Readout diagnosis and inference. Matched ridge classifiers are fitted separately to smooth and spiking final-layer features, using the same source-training labels and validation penalty grid. Encoders remain frozen; no target labels enter fitting. We report balanced accuracy (BA; chance 25%), averaging seeds within participants before 10,000 participant bootstrap resamples. Contrasts use paired 95% intervals. The primary contrast is calibrated SAST minus matched ordinary training. Readout diagnostics were planned secondary analyses; this study was not preregistered. A post-study check repeats the readout analysis with a single-block depthwise LIF encoder and four temporal bins (same nine participants, five seeds, source search).

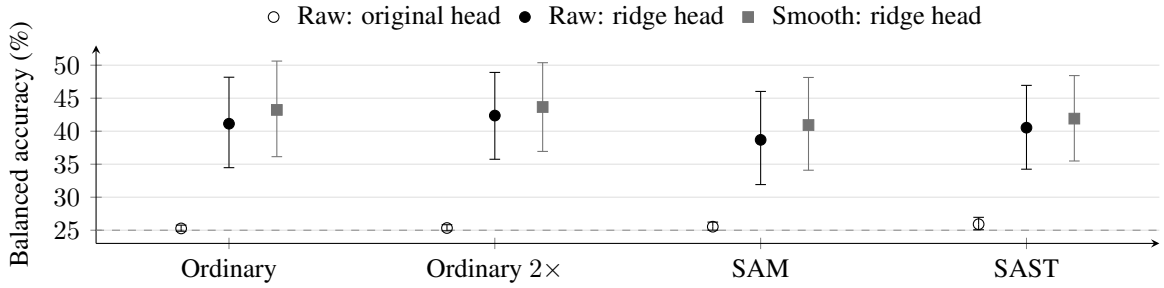


Figure 1: Source-only readout refitting recovers decoding across all four training conditions. Points are nine-participant means; bars are 95% bootstrap intervals. The dashed line is chance. Both ridge heads use identical source splits and tuning budgets; smooth original-head scores are in Table 1.

2 Results

Table 1: Session 2 BA (%), mean [95% participant interval]. Calibration uses source labels only; – indicates an inapplicable mode.

Decoder / training	Nonspiking	Raw spikes	Calibrated spikes
Ordinary	41.2 [34.2, 48.8]	25.3 [24.8, 25.8]	32.7 [28.9, 36.8]
Ordinary, 2× updates	42.5 [35.4, 49.8]	25.3 [24.9, 25.8]	33.5 [30.2, 36.7]
Same-batch SAM	37.6 [30.9, 44.8]	25.5 [25.0, 26.2]	32.3 [27.9, 37.2]
Independent SAST	39.5 [33.2, 46.0]	25.9 [25.1, 27.0]	33.0 [29.1, 37.9]
Direct hard-forward	–	43.0 [35.9, 50.2]	43.0 [35.9, 50.3]
EEGNet reference	54.1 [43.8, 64.2]	–	–
FBCSP-style reference	61.8 [52.9, 70.1]	–	–

Conversion and the primary comparison. The original study's 270 main fits and 216 tuning fits completed. Raw conversion reduces mean BA to near chance in every smooth-training condition (Table 1). Source threshold calibration partly recovers accuracy. Calibrated SAST minus matched ordinary training is -0.42 percentage points (pp), 95% CI $[-2.88, 2.67]$; the artifact-free sensitivity gives -0.32 pp $[-2.87, 2.87]$. The interval spans zero and effects in both directions: the study does not demonstrate SAST superiority or establish equivalence.

Recovery with a frozen encoder. Source-fitted raw-spike probes reach 38.69–42.37% BA (Figure 1). For SAST, refitting raises BA from 25.91% to 40.53%: $+14.62$ pp $[8.26, 21.37]$. Refitting also helps smooth features ($+2.44$ pp $[1.33, 3.51]$), so the entire raw-spike gain cannot be attributed uniquely to conversion repair. Comparing matched probes, raw SAST remains 1.37 pp below smooth SAST (raw minus smooth: $[-2.03, -0.71]$); all four methods retain a 1.30–2.25 pp deficit. Recovery coexists with residual loss of linear decodability. The post-study FBCSP-style reference reaches 61.81% BA, exceeding EEGNet by 7.66 pp $[3.67, 11.91]$ on the same test split.

Post-study architecture check. A second encoder yields 27.44% raw BA and 37.55% after source-only refitting (gain $+10.11$ pp $[5.86, 14.29]$); recovery exceeds matched smooth refitting by $+7.09$ pp $[2.01, 12.21]$.

3 Discussion

For this offline task, frozen-readout failure overstates the loss of linearly decodable EEG information. Before collecting target labels, refit matched smooth and spiking readouts using source labels. Recovery does not demonstrate unchanged neural computation or a causal effect of flatness. Converted models remain below both EEGNet and the classical reference. Nine participants, one dataset, and two LIF encoders limit generalization. No online-control, clinical, or hardware-energy benefit is established.

Data, code, and related work. This is secondary analysis of public EEG; no new participants were recruited. Related work includes the author's SAST preprints [1, 2] and a submitted, unpublished UniReps manuscript on representational fidelity. The EEG evaluation and frozen-feature readout diagnosis are separate contributions. Code and outputs: `https://github.com/PluvioXO/ai4bci2026-readout`.

References

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